

Once the above command is executed, you will be asked to provide some basic information. The output will be very similar to:

```
Generating a 1024 bit RSA private key
.....+++++
.....+++++
writing new private key to '/etc/vsftpd/vsftpd.pem'
-----
You are about to be asked to enter information that will be
incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished
Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:IN
State or Province Name (full name) [:Maharashtra
Locality Name (eg, city) [Default City]:Pune
Organization Name (eg, company) [Default Company
Ltd]:MyTestOrganizationLtd
Organizational Unit Name (eg, section) [:Information
Technology
Common Name (eg, your name or your server's hostname) [:My
Test FTP Server
Email Address [:mandar.shinde2007@gmail.com
```

The vsftpd configuration part

After generating the SSL certificate, we need to instruct vsftpd to use that SSL certificate to carry out the encryption process. Just like many services, vsftpd has its own configuration file, *vsftpd.conf*, which is located in */etc/vsftpd/vsftpd.conf* for Red Hat-based systems and */etc/vsftpd.conf* in Debian-based systems.

Now, let us edit the configuration file as per our requirements. You might need to find out the lines, or add them if they do not pre-exist.

Step 1: Turn on SSL.

We would like to enable encryption not only for data transfer, but also for authentication process. For this, you can edit following lines as:

```
ssl_enable=YES
allow_anon_ssl=NO
force_local_data_ssl=YES
force_local_logins_ssl=YES
```

Step 2: Mention the certificate and key file location.

```
rsa_cert_file=/etc/vsftpd/vsftpd.pem
rsa_private_key_file=/etc/vsftpd/vsftpd.pem
```

Step 3: Enable TLS. TLS is considered to be more secure than SSL so we would definitely like to use it whenever required.

```
ssl_sslv2=YES
ssl_sslv3=YES
```

Step 4: This includes other basic configurations. To allow all the local users added to the system to use FTP service, edit the following line:

```
local_enable=YES
```

To prevent anonymous logins, edit the following line:

```
anonymous_enable=NO
```

To accept FTP write commands, edit the following line:

```
write_enable=YES
```

With this setting, only a local user can access the FTP server and can issue write commands. But if you want to preserve the individuality of the users and their contents, you can set up a *chroot* jail for the users, so that users are bound to work in their home directories and are not permitted to access any files outside them.

```
chroot_local_user=YES
```

To enable logging of the transfers carried out, edit the following lines:

```
xferlog_enable=YES
xferlog_std_format=YES
xferlog_file=/var/log/ftp/xferlog
```

Add the vsftpd service to start up

With the configuration done, you will have to restart the service so that the changes incorporated can take effect:

```
service vsftpd restart
```

By default, after a fresh installation of any package, the service associated with that package is disabled on every run level. This indicates that you will have to manually restart the service after the operating system switches from one run level to another. In simple words, after every reboot or system startup, you will have to start the service manually.

You can verify this by issuing the *chkconfig* command as follows:

```
chkconfig --list vsftpd
```